CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

May 2012

EC 208 Analog Electronic Circuits

Date: 17.05.2012, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

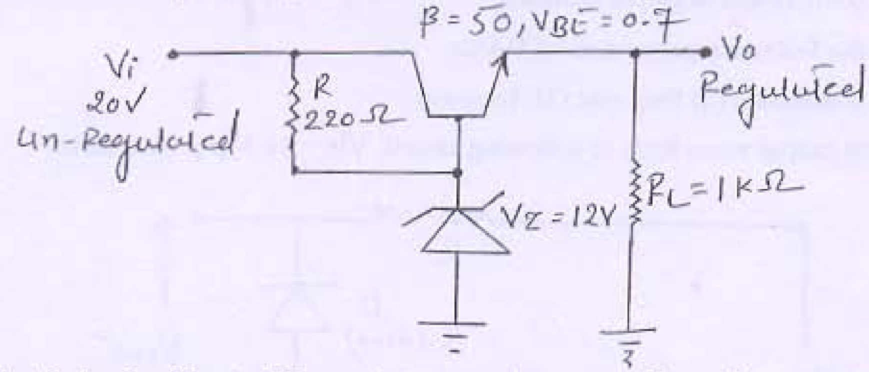
Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Calculate the effective resistance seen looking into the primary of a 15:1 transformer connected to an 8 ohm load. **[02]**

(b) Calculate the output voltage and zener current in the regulator circuit. **[05]**



Q - 2 (a) What is the significant difference between voltage amplifier and power amplifier? **[02]**

(b) Draw the circuit of an emitter coupled differential amplifier and explain why the $CMRR \to \infty$ for a symmetrical circuit with $R_e \to \infty$. **[06]**

(c) Explain the counter type A to D Converter. **[06]**

OR

Q - 2 (a) Compare the power amplifier classes in terms of their efficiency. **[02]**

(b) Draw the equivalent circuit from which to calculate common mode gain and difference gain for the emitter coupled differential amplifier. **[06]**

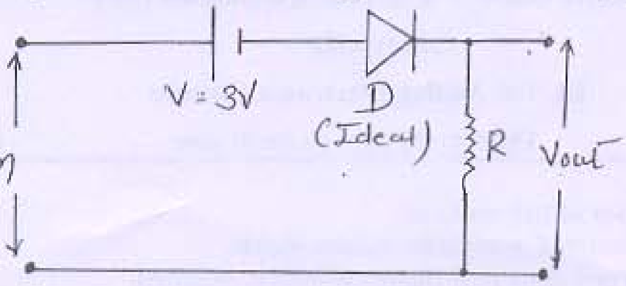
(c) Explain the successive approximation type ADC. **[06]**

Q - 3 (a) What are the four possible topologies of a feedback amplifier? Identify the output signal X_o and the feedback signal X_f for each topologies **[04]**

(b) State the three fundamental assumptions which are made in order that the expression $A_f = A/(1 + A\beta)$ be satisfied exactly. **[04]**

(c) Explain weighted resistor type DAC [03]

(d) Draw the output wave form of following circuit. Vin = 10 V p-p Sine wave. [03]



OR

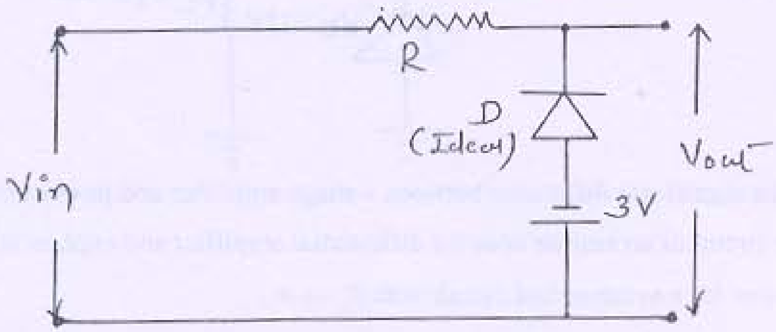
Q – 3 (a) Define desensitivity D. for a large value of D what is A_f? What is a significance of this result? [04]

(b) Define positive and negative feedback. [04]

(c) Define the following parameters of DACs: [03]

(1) Resolution (2) Step size (3) Accuracy

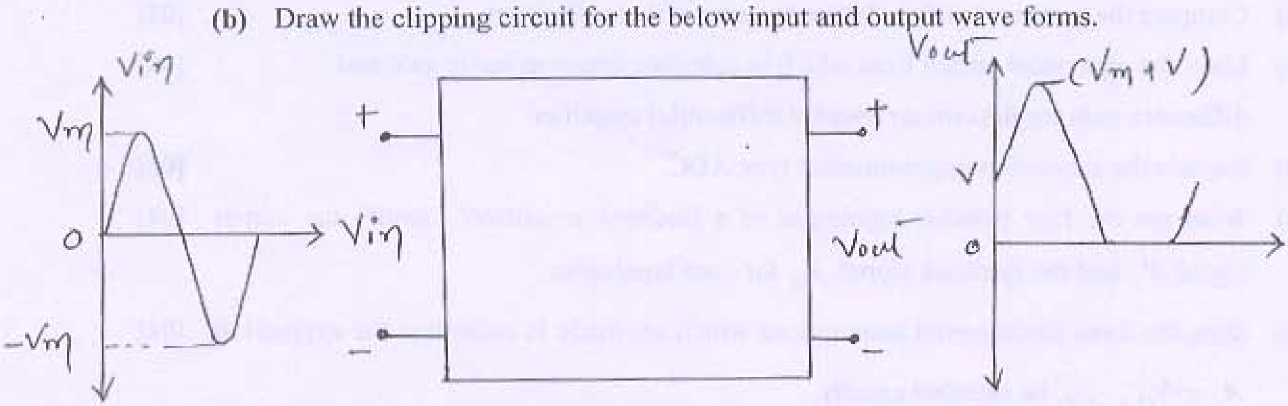
(d) Draw the output wave form of following circuit. Vin = 10 V p-p Sine wave. [03]



SECTION – II

Q – 4 (a) What is the condition for the sustained oscillation? [02]

(b) Draw the clipping circuit for the below input and output wave forms. [05]

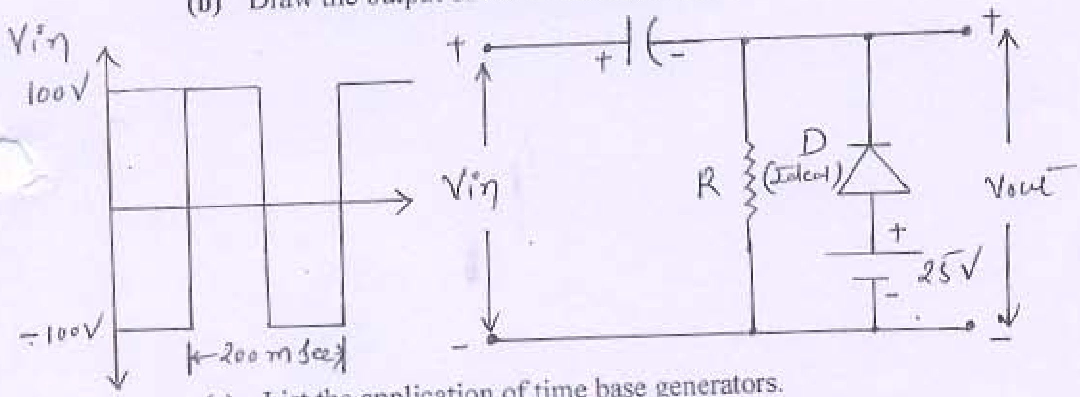


Q – 5 (a) Explain the basic principle of RC phase shift oscillator. [02]

(b)	Explain hartly oscillator with necessary circuit diagram.	[04]
(c)	Explain the V-I Characteristic of tunnel diode.	[04]
(d)	Explain general features of a time base signal.	[04]

OR

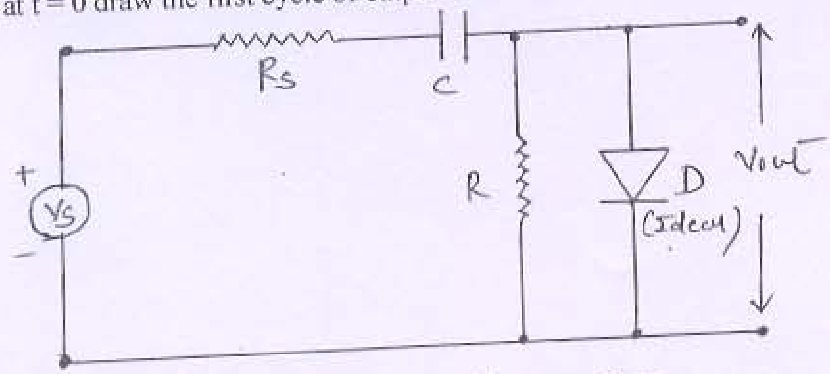
Q – 5 (a)	Can a negative feedback amplifier work as an oscillator? If yes, how? If no, why?	[02]
(b)	Draw and explain the wein bridge oscillator.	[04]
(c)	Explain the unijunction transistor.	[04]
(d)	Explain the basic principles of miller and bootstrap time base generators.	[04]
Q – 6 (a)	Explain Clamping circuit theorem.	[03]
(b)	Draw the output of the following circuit.	[04]



(c)	List the application of time base generators.	[02]
(d)	Explain a simple current sweep in time base generator.	[05]

OR

Q - 6 (a) In the circuit below Rs = Rf = 50 ohm, R = 20 Kohm and C = 2 micro-ferade. A symmetrical square wave signal of amplitude 20 volt and frequency 5 KHz is applied at t = 0 draw the first cycle of output waveform. [07]



(b)	List the methods of generating a time base wave forms.	[02]
(c)	Explain a transistor current time base generator.	[05]

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